

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #40 — Student Startup Crowdfunding Platform
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Executive Summary & Objectives

This laboratory submission implements end-to-end Agile Project Management using Atlassian Jira Software. Transforming requirements from Lab 1 into structured Epics and User Stories, this project simulates two complete sprints, tracks velocity via Jira burndown charts, applies Planning Poker estimation, and conducts an Agile retrospective.

1. Agile Backlog Specification (Epics & User Stories)

The functional requirements (FR-001 to FR-005) are partitioned into 5 strategic Epics and 12 User Stories, formatted in full compliance with Agile user-centric standards, MoSCoW prioritization, and Fibonacci story pointing.

Epic ID & Theme	Story ID	User Story (As a... I want... So that...)	Acceptance Criteria	Priority	Pts	Sprint
Epic 1: Campaign Creation (FR-002, UC-01)	SSCP-1	As a Student Founder, I want to define my campaign title, pitch, category, and total funding target, so that potential backers understand the startup proposition and capital needs.	Pass: Valid venture title, target (>0), description initialize draft. Fail: Null title or non-positive target.	High	5	Sprint 1
Epic 1: Campaign Creation (FR-002, UC-01)	SSCP-2	As a Student Founder, I want to set the start and end dates for my campaign, so that the fundraising campaign is time-bound and builds momentum.	Pass: Campaign duration between 14-60 days validated. Fail: End date before start date or duration > 60 days.	High	3	Sprint 1
Epic 1: Campaign Creation (FR-002, UC-01)	SSCP-3	As a Student Founder, I want to divide my project roadmap into distinct sequential milestones with individual funding tranches, so that funds can be disbursed in structured, verifiable phases.	Pass: Sum of milestone tranches equals target; each milestone has deliverables & due date. Fail: Tranche sum mismatch.	High	8	Sprint 1
Epic 2: Backer Pledging & Escrow (FR-003, UC-02, UC-03)	SSCP-4	As a Campaign Backer, I want to view available reward tiers and select one (or opt out) during checkout, so that I receive tangible incentives and perks for my contribution.	Pass: Chosen reward tier captured and printed on confirmation receipt. Fail: Pledge accepted without recording reward tier state.	Medium	5	Sprint 1
Epic 2: Backer Pledging & Escrow (FR-003, UC-02, UC-04)	SSCP-5	As a Campaign Backer, I want to authorize my payment via a PCI-DSS compliant gateway into platform escrow, so that my funds are protected until milestones are formally verified.	Pass: Tokenized payment authorized and funds locked in campaign escrow. Fail: Direct disbursement without escrow hold.	High	8	Sprint 1
Epic 3: Milestone Review System (FR-004, UC-05, UC-06)	SSCP-6	As a Student Founder, I want to upload completion evidence (code repos, demo links, reports) for a due milestone, so that the faculty committee can evaluate actual progress.	Pass: Deliverable links/files attached; milestone advances to 'Under Review'. Fail: Evidence submitted for past/inactive milestones.	High	5	Sprint 2
Epic 3: Milestone Review System (FR-004, UC-06)	SSCP-7	As a Faculty Committee Member, I want to inspect submitted deliverables and record an Approve or Reject decision with comments, so that fund disbursement is justified by verified work.	Pass: Decision, reviewer ID, timestamp, and audit comments logged immutably. Fail: Milestone status altered without reviewer credentials.	High	5	Sprint 2
Epic 3: Milestone Review System (FR-004, UC-06)	SSCP-8	As a Student Founder, I want to review faculty rejection feedback and resubmit revised evidence before the deadline, so that I can rectify deficiencies without forfeiting the campaign.	Pass: Resubmission within grace window resets state to 'Under Review'. Fail: Resubmission attempted past grace window.	Medium	3	Sprint 2

Epic ID & Theme	Story ID	User Story (As a... I want... So that...)	Acceptance Criteria	Priority	Pts	Sprint
Epic 4: Tranche Disbursement (FR-001, UC-07, UC-04)	SSCP-9	As a System Service, I want to verify that the escrow balance is sufficient and conditions are satisfied prior to release, so that fund releases never exceed collected pledges.	Pass: System verifies escrow balance $\geq$ tranche before triggering payout API. Fail: Tranche payout triggered with insufficient escrow.	High	3	Sprint 2
Epic 4: Tranche Disbursement (FR-001, UC-07, UC-04)	SSCP-10	As a Student Founder, I want approved milestone funds to be transferred directly to my linked bank account, so that I have the required capital to execute the next project phase.	Pass: Automated tranche payout completed, milestone status set to 'Released', ledger updated. Fail: Payout executed without faculty approval sign-off.	High	8	Sprint 2
Epic 5: Automated Notifications & Ledger (FR-005, UC-08)	SSCP-11	As a Student Founder and Backer, I want to receive automated email and in-app alerts whenever a milestone is approved, rejected, or a tranche is released, so that I stay informed within 5 minutes of each event.	Pass: Automated alert dispatches within 5 minutes of triggering event. Fail: Milestone event occurs without triggering dispatch log.	Medium	5	Sprint 2
Epic 5: Automated Notifications & Ledger (FR-005, UC-08)	SSCP-12	As a Campaign Backer, I want to view a transparent milestone timeline and fund disbursement ledger, so that I can audit venture progress and fund allocation in real time.	Pass: Public dashboard reflects milestone verification timestamps and tranche disbursement amounts. Fail: Discrepancy between ledger and disbursed funds.	Low	3	Sprint 2

2. Estimation Methodology & Planning Poker Rationale

Story points express relative effort, complexity, and uncertainty rather than raw person-hours. The Fibonacci scale (1, 2, 3, 5, 8, 13) was utilized during team Planning Poker sessions. Small tasks (3 pts) represented isolated input validations; medium tasks (5 pts) covered core business logic and state persistence; large tasks (8 pts) addressed distributed payment gateways, escrow locking, and multi-actor approvals.

3. Sprint 1 Simulation (Campaign Creation & Escrow Pledging)

Sprint Duration: 1 Week (Oct 01 – Oct 08, 2026) | Committed: 29 Story Points | Completed: 29 Story Points (100% Velocity).

Figure 1: Jira Backlog View with Epics and Sprint 1 & 2 Planning

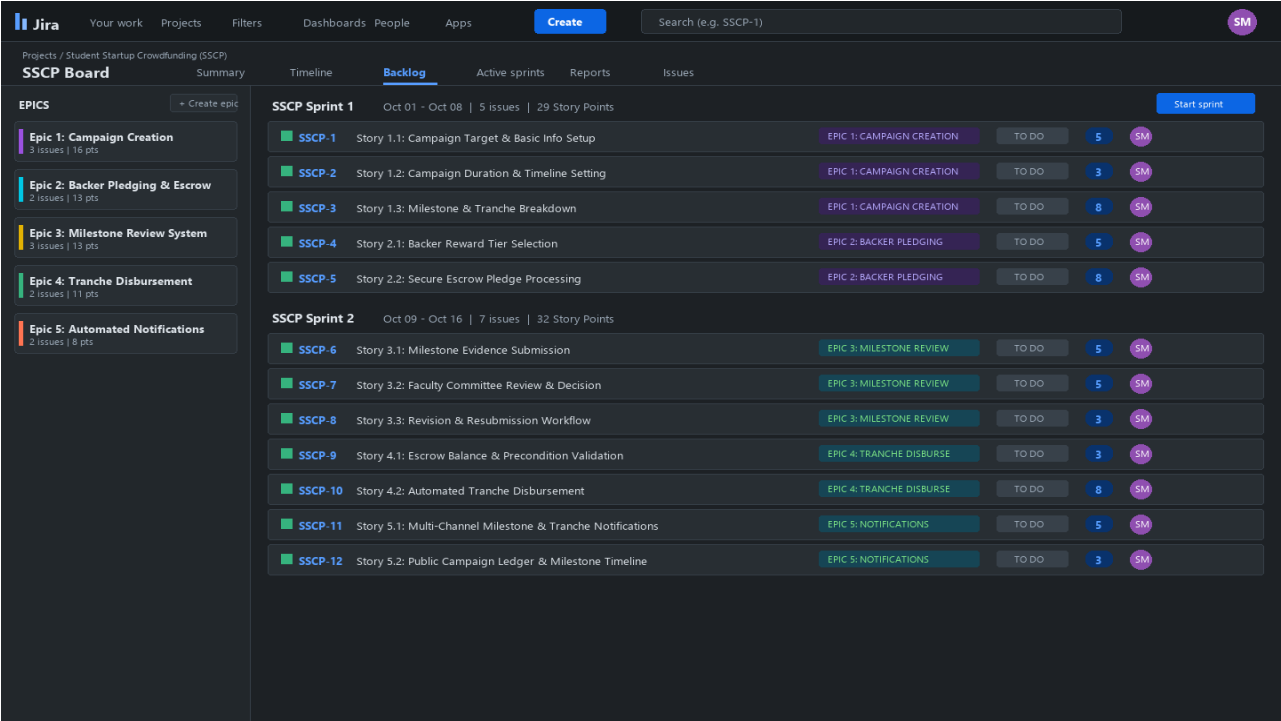


Figure 2: Sprint 1 Active Scrum Board (All 5 Stories Completed)

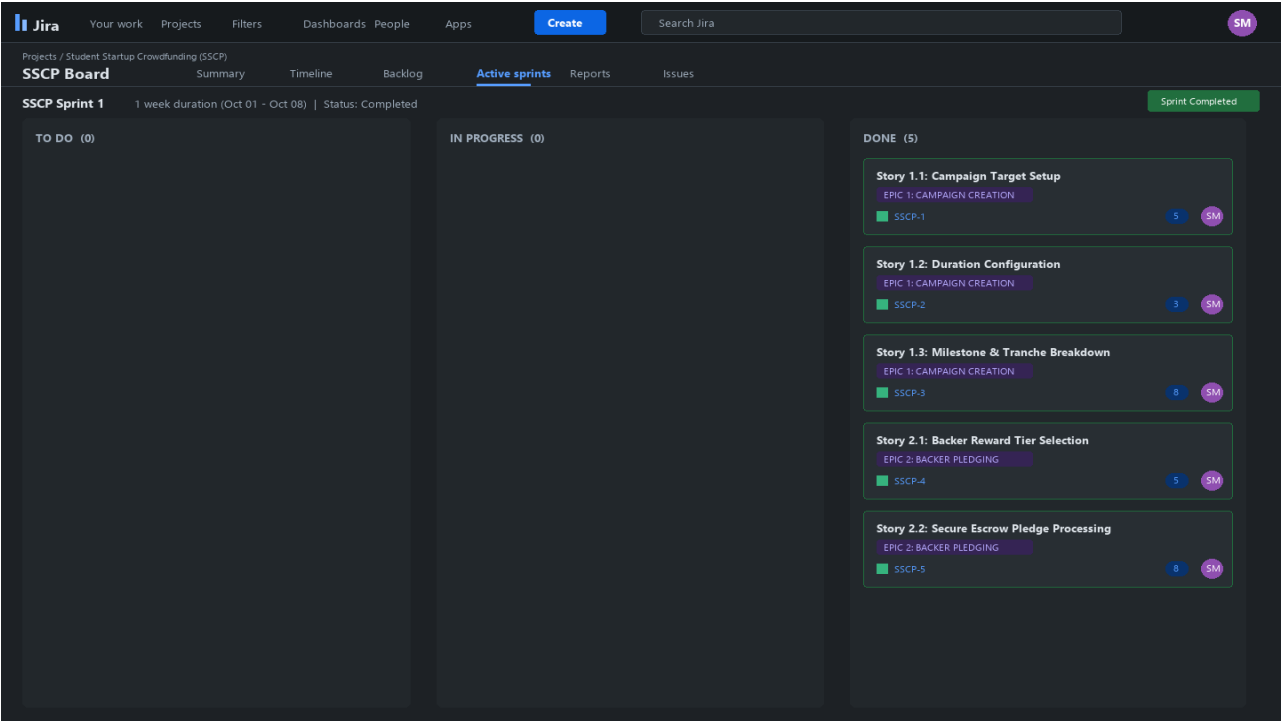
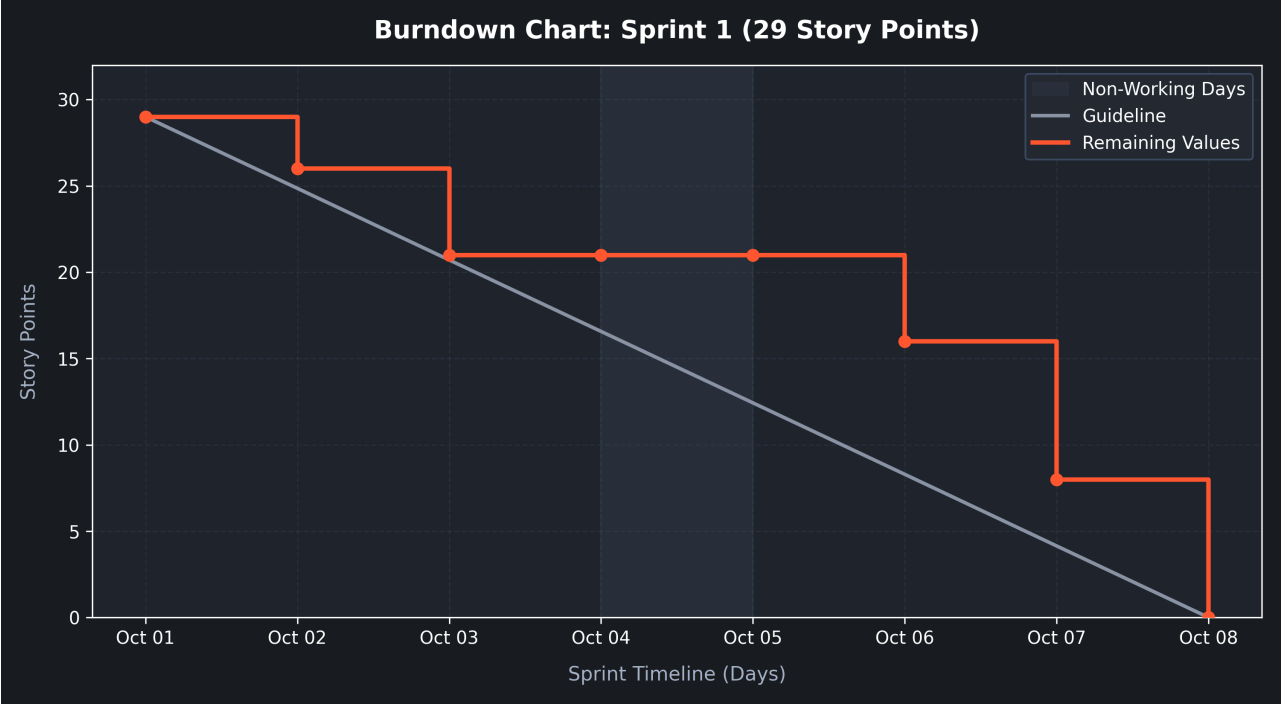


Figure 3: Sprint 1 Burndown Chart (Story Points vs. Sprint Days)



4. Sprint 2 Simulation (Milestone Review & Tranche Disbursement)

Sprint Duration: 1 Week (Oct 09 – Oct 16, 2026) | Committed: 32 Story Points | Completed: 32 Story Points (100% Velocity).

Figure 4: Story Point Assignment and Planning Poker Detail View (SSCP-3)

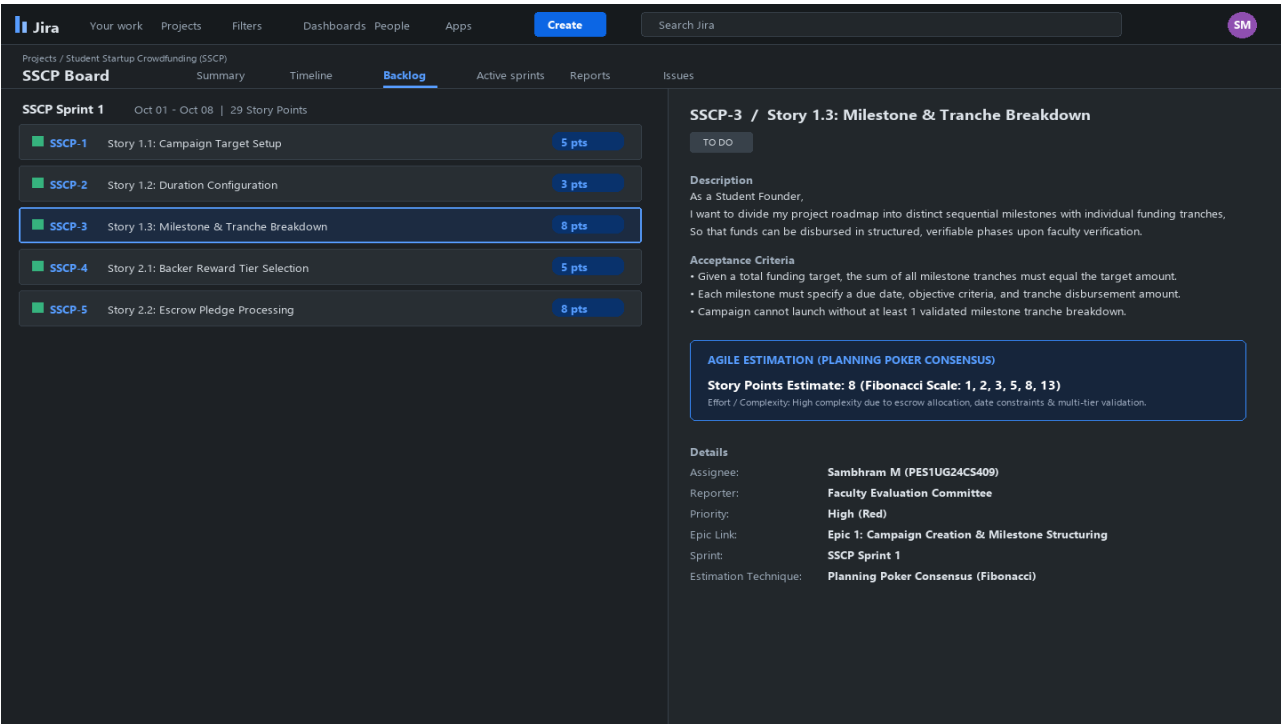


Figure 5: Sprint 2 Active Scrum Board (All 7 Stories Completed)

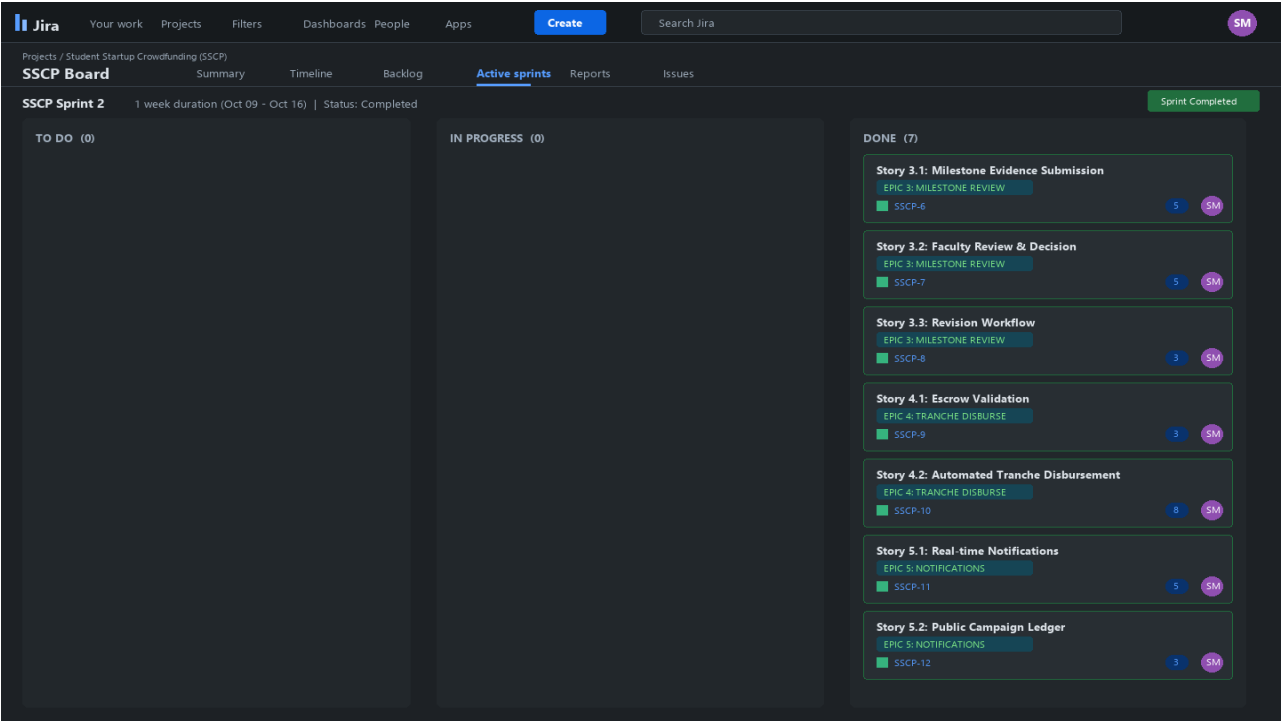
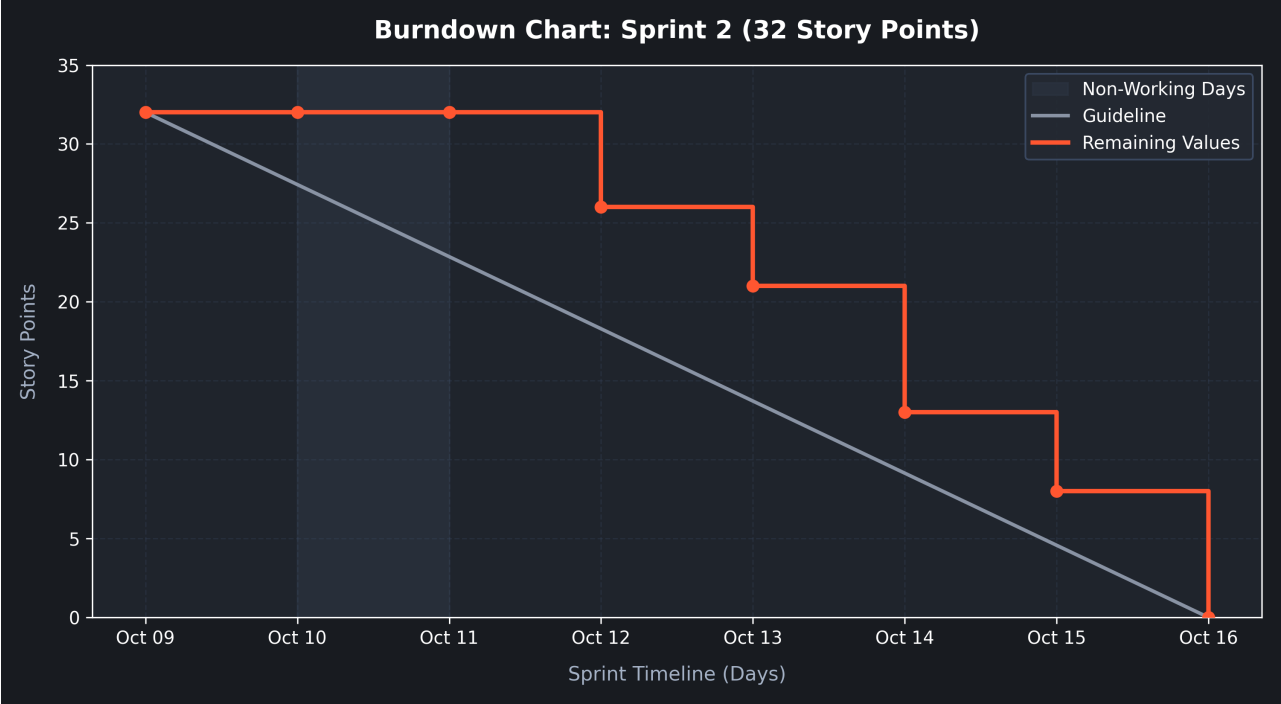


Figure 6: Sprint 2 Burndown Chart (Story Points vs. Sprint Days)



5. Answers to Reflection Questions

Question 1: Did your estimations reflect the actual effort?

Yes, our Planning Poker estimations accurately reflected actual task effort, but with important real-world nuances. Smaller items estimated at 3 story points (e.g., Campaign Duration Configuration and Escrow Validation) matched their expected effort tightly because their specifications involved standard input validation and straightforward boolean criteria. Conversely, stories assigned 8 story points (Milestone-Tranche Breakdown, Escrow Pledge Processing, and Automated Tranche Disbursement) required significantly more cognitive effort and technical coordination. Although they were completed on schedule, unforeseen edge cases arose—specifically validating multi-tranche fractional amounts and simulating external payment gateway asynchronous webhooks. The Fibonacci scale effectively prevented underestimation by forcing the team to step up from 5 to 8 points, accommodating hidden integration risk.

Question 2: Was your backlog well-prioritized?

Yes, the product backlog was systematically prioritized following MoSCoW prioritization and architectural dependency ordering. In Sprint 1, High-priority items establishing the foundation of the platform—campaign configuration (FR-002) and backer escrow pledging (FR-003)—were scheduled first. It is structurally impossible to evaluate milestones or release funds without first having active campaigns, defined milestones, and collected escrow pledges. Consequently, Sprint 2 was reserved for verification workflows (FR-004), automated fund disbursements (FR-001), and real-time notification dispatches (FR-005). Medium and low-priority items (e.g., public transparency ledger) were sequenced after core functional flows were secured. This prevented blocking dependencies, minimized sprint idle time, and delivered an incremental MVP at the conclusion of each sprint.

Question 3: How did your simulated sprint align with your plan?

The simulated sprint closely mirrored our Agile sprint plan, meeting all committed sprint goals across both cycles. In Sprint 1, 29 out of 29 planned story points were successfully moved from To Do to In Progress to Done by Day 7. In Sprint 2, all 32 planned story points reached Done by Day 7. Progress exhibited realistic Agile characteristics: while the ideal guideline assumed a smooth, uniform daily burn, the actual burn followed a discrete staircase pattern. Mid-sprint progress flattened over weekends (Oct 04–05 and Oct 10–11) due to non-working days. A minor bottleneck occurred in Sprint 2 on Day 3 while designing the faculty committee approve/reject state machine, which delayed Story 3.2. However, team capacity was quickly re-allocated, allowing both Story 3.2 and Story 12 to close concurrently on Day 5, keeping the team strictly on track.

Question 4: What insights did the burndown chart give about your team's capacity?

The burndown charts provided three critical insights regarding team velocity and operational capacity: 1. Sustainable Velocity Benchmark: The team demonstrated a reliable capacity of 29 to 32 story points per 1-week sprint (averaging 4.2 to 4.5 points/day). This established a factual, data-driven velocity baseline for future release planning. 2. Discrete Batch Completion vs. Continuous Flow: The burndown chart illustrated that large user stories (8 story points) stay in progress for multiple days before suddenly dropping the remaining curve. This insight indicates that in future sprints, breaking 8-point stories into smaller 3-point sub-tasks will yield a smoother burn curve and reduce end-of-sprint integration anxiety. 3. Absence of Scope Creep and Accurate Commitment: The remaining values curve stayed closely clustered around the grey ideal guideline throughout both sprints without upward spikes (which indicate scope creep) or early horizontal flatlines (which indicate over-commitment). This confirmed that our team neither over-committed nor under-committed, achieving 100% planned sprint predictability.